

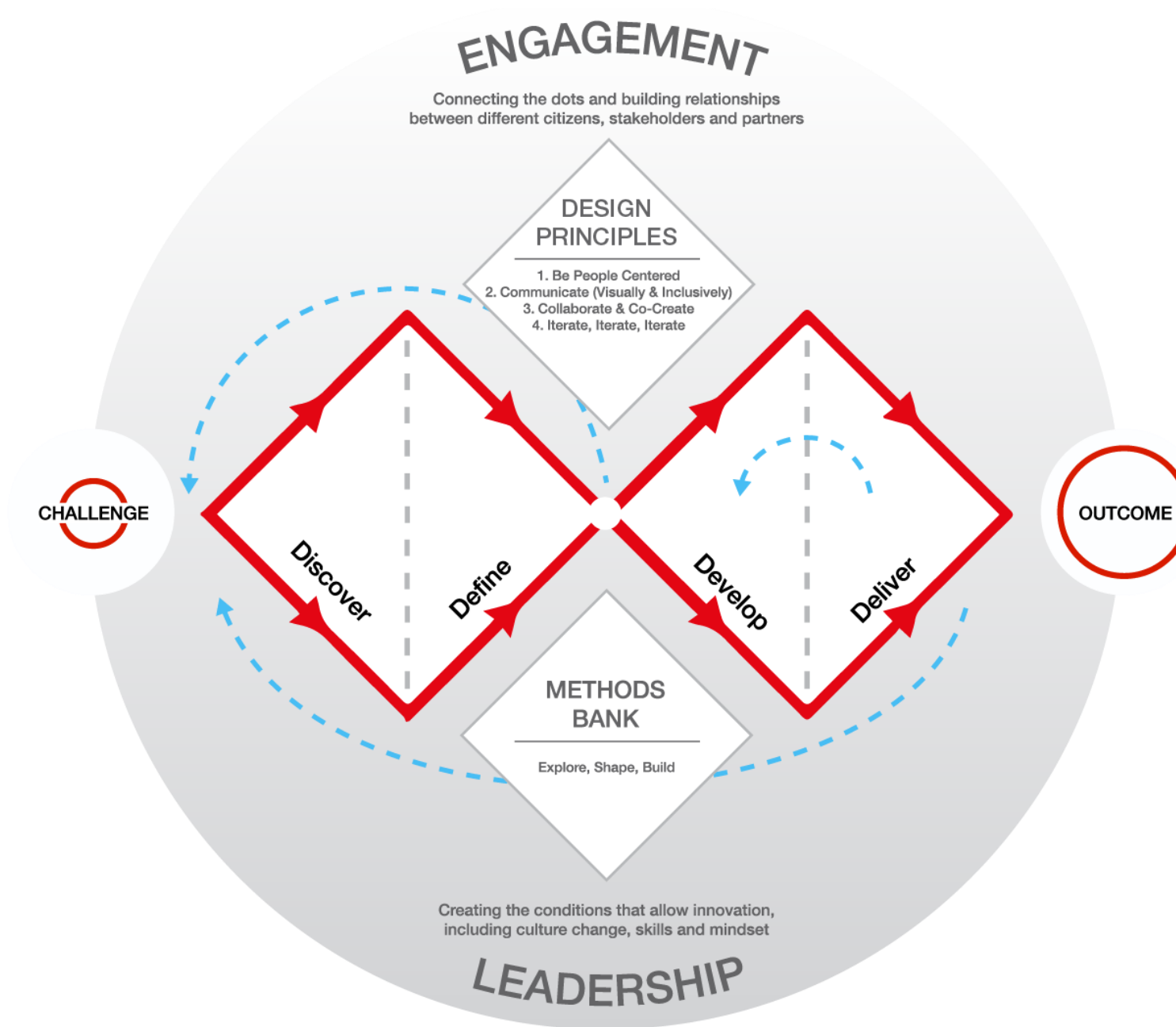
Human-centred AI as a Design Process

Part 2: Prototyping human-centred AI systems

Dr Georgia Panagiotidou

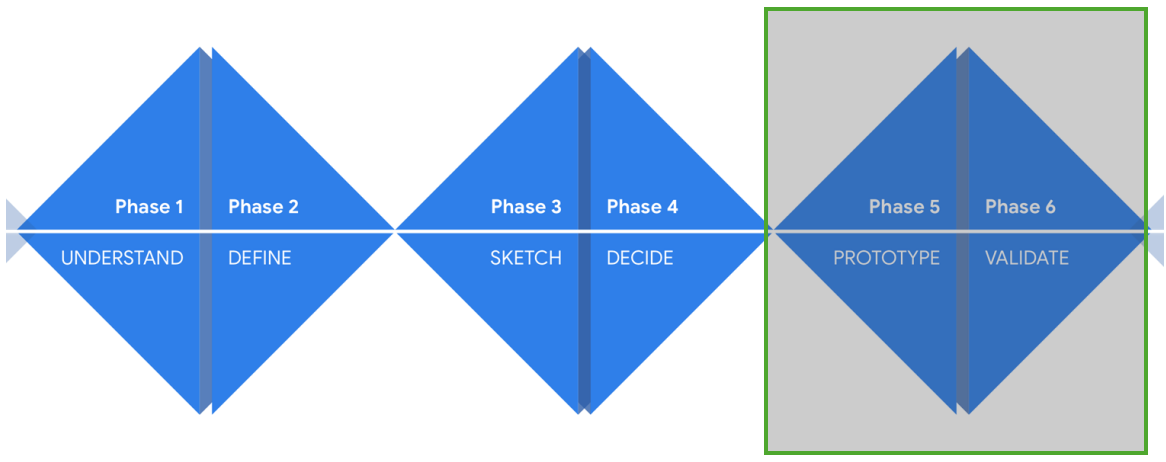
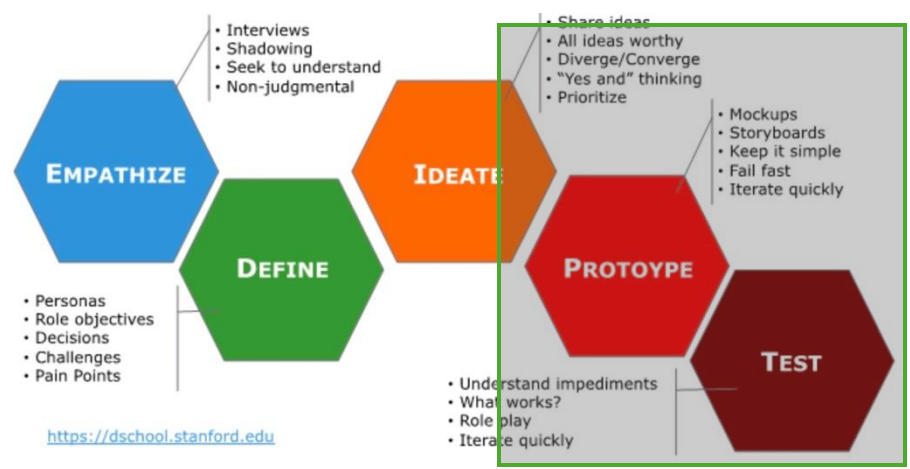
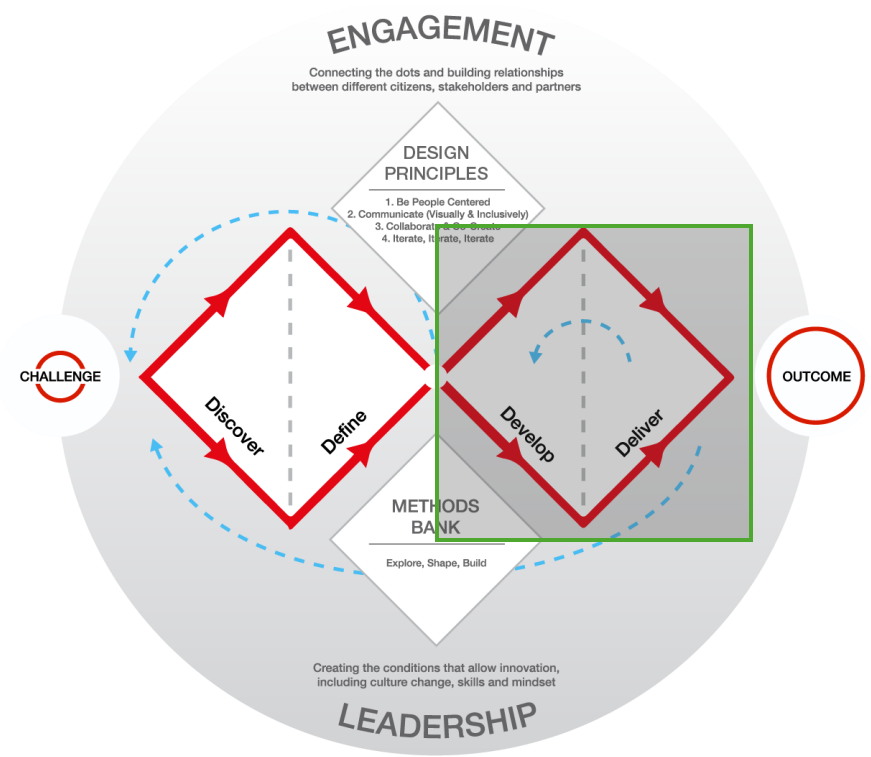
This week we will:

- understand the basic goal of prototyping
- understand key conceptual ideas in prototyping
- explore examples of AI specific prototyping techniques



Today's focus

- Prototype

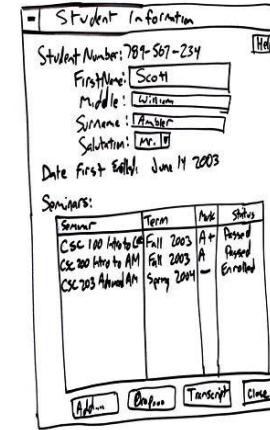


What are prototypes?

“A prototype is one manifestation of a design that allows stakeholders to interact with it and to explore its suitability;
it is limited in that a prototype will usually emphasize one set of product characteristics and de-emphasize others”

Sharp, Preece, Rogers. “Interaction Design.”

What are prototypes?



A hand-drawn paper prototype of a student information form. The form is titled "Student Information" and includes fields for "Student Number: 789-867-234", "First Name: Scott", "Middle: William", "Surname: Zander", "Salutation: Mr.", and "Date First Set: June 14 2003". Below this is a table titled "Semesters:" with columns for "Semester", "Term", "Avg", and "Status". The table contains three rows of data. At the bottom of the form are four buttons: "Add...", "Refresh", "Transcript", and "Close".

Semester	Term	Avg	Status
CSC 100 Intro to CS	Fall 2003	A+	Passed
CSC 200 Intro to AI	Fall 2003	A	Passed
CSC 203 Advanced AI	Spring 2004	-	Enrolled



.... can be anything from:

- a paper-based storyboard through to a complex piece of software;
- from a cardboard mock-up to a moulded or pressed piece of metal.

The purpose is to help users envision the important parts of interactions with the newly designed systems

Why prototype?

Prototypes can:

- help you **think through** and **understand the possibilities** in your design space
- help you **test assumptions** about **users**, the **experience**, or **viable technologies**.

Why prototype human-centred AI?

- To align AI system behaviour with user expectations
- To catch usability issues early
- To explore ethical, trust, and explainability concerns before implementation

Traditional AI

Model-centric

Data first

Evaluate by
accuracy

Human-Centred AI

User-centric

People first

Evaluate by usability, trust

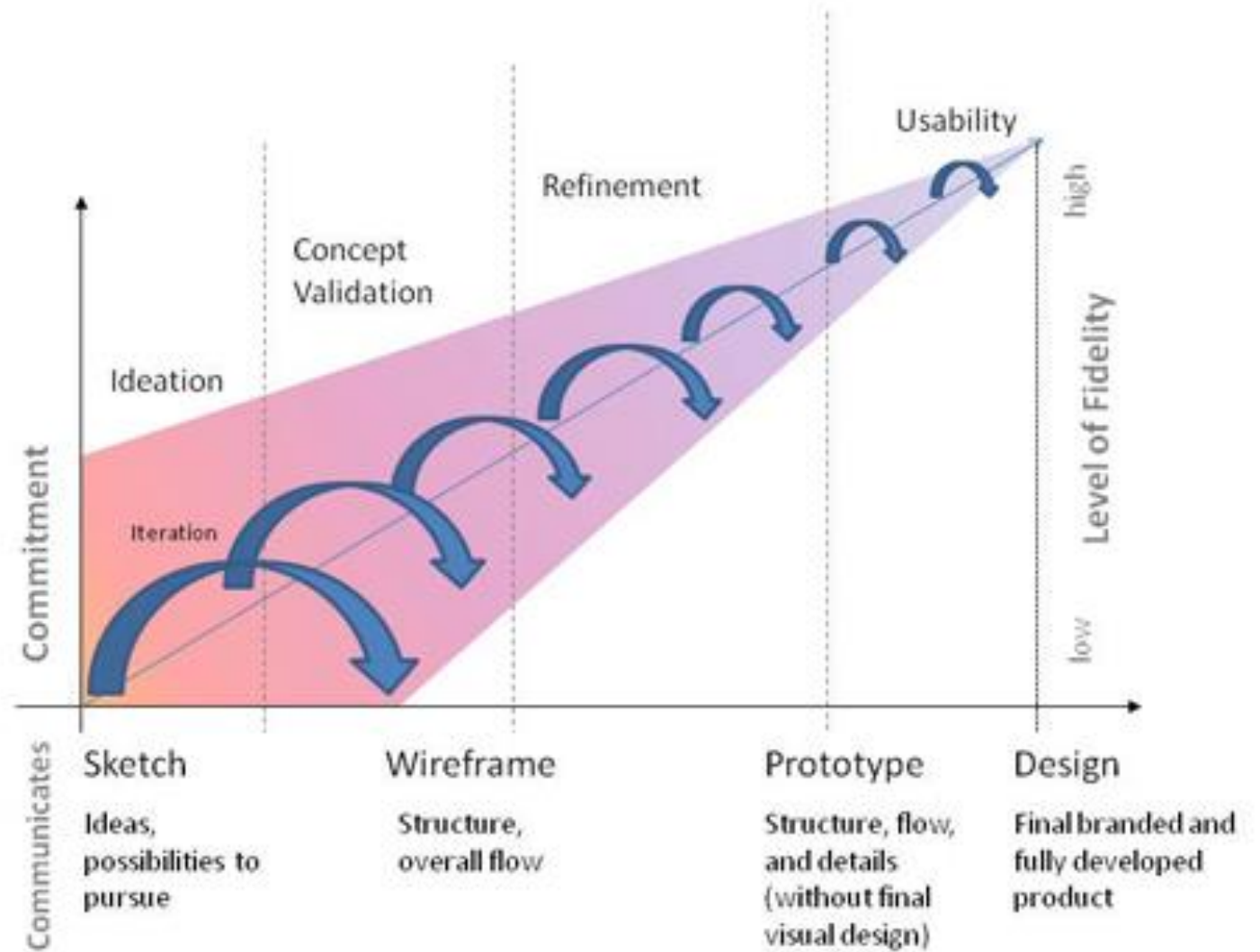
Types of prototypes

Low & High Fidelity

Type	Advantages	Disadvantages
<u>Low-fidelity</u>	- Lower "development" cost - Easier to evaluate multiple concepts - Capture contexts of use, market needs - Less constrained (conceptual models) - Concept communication tool - Explore high-level design & layout options	- No error checking - Little/no detail - Difficult / poorly suited for usability assessment - Navigation limitations - Content may not fit layout - Can't test dynamic layout
<u>High-fidelity</u>	- Partially functional - Interactive - Contains content: effective preview of layout - Usability test w/ target users, identify UX bugs - Look & feel resembles final product "Living specification" - Use as marketing and sales tool	- Time & resource intensive to develop and change - Not effective for requirements gathering or early-stage ideation - May confuse stakeholders prototype vs final product

adapted from 11.2, Beyond Interaction Design

Sketches, Wireframes & Prototypes

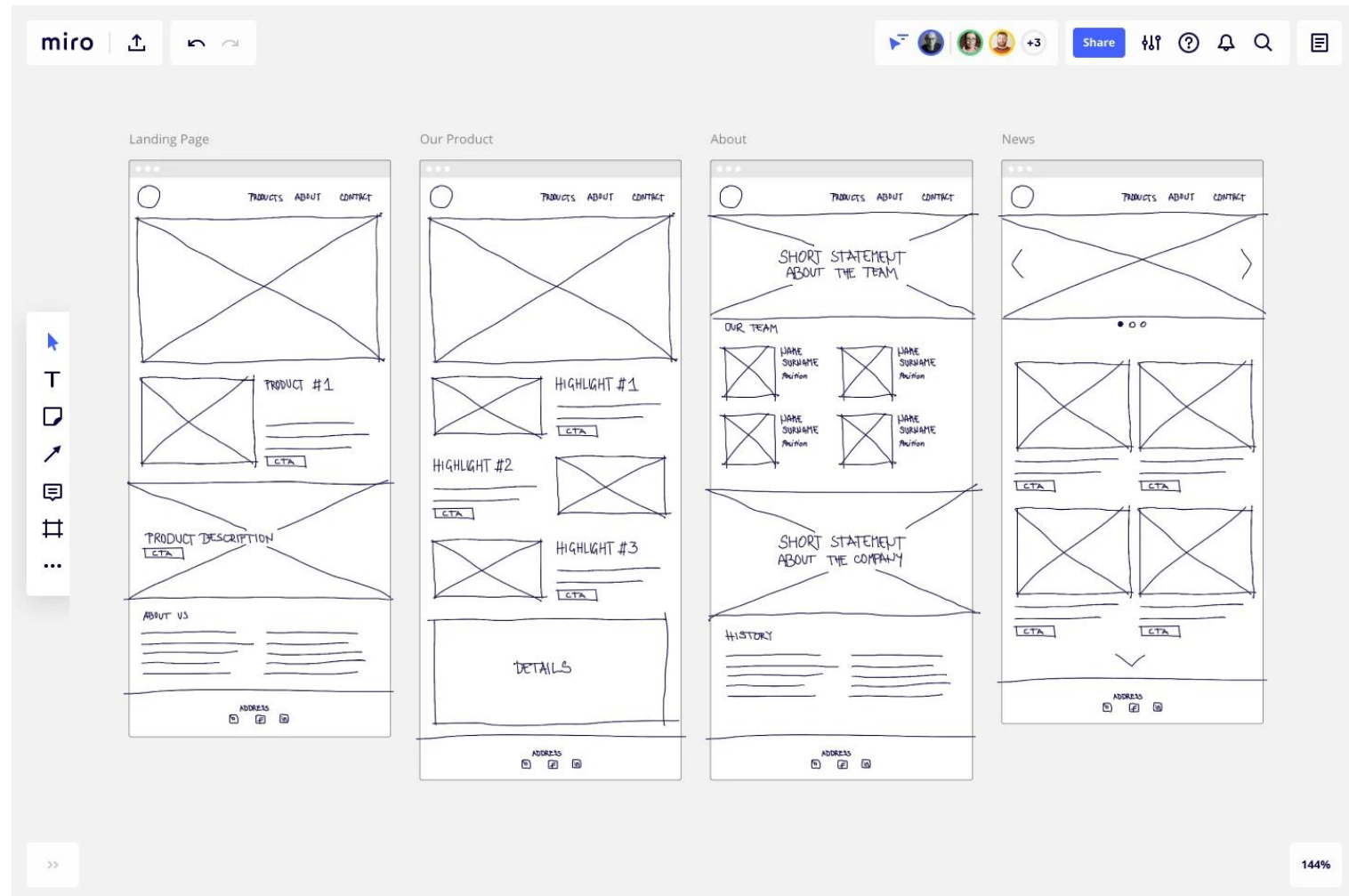


<https://www.uxmatters.com/mt/archives/2010/05/sketches-and-wireframes-and-prototypes-oh-my-creating-your-own-magical-wizard-experience.php>

Low-fidelity Prototypes

Wireframes

- Wireframing is a process where designers draw overviews of interactive products to establish the structure and flow of possible design solutions.
- Wireframes are primarily tools for collaboration toward making better prototypes and products *faster*.



Wiki

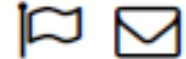
[Welcome Peldi](#) | [History](#) | [Preferences](#) | [Administration](#) | [Log Out](#) 

Page Operations

Browse Space

Add Content

Main



Welcome to our Wiki!

Added by Peldi, last edited by Peldi on Oct 17, 2013

Welcome to our awesome wiki, a place for us to share knowledge yadda yadda yadda.
You can share images like the one below:

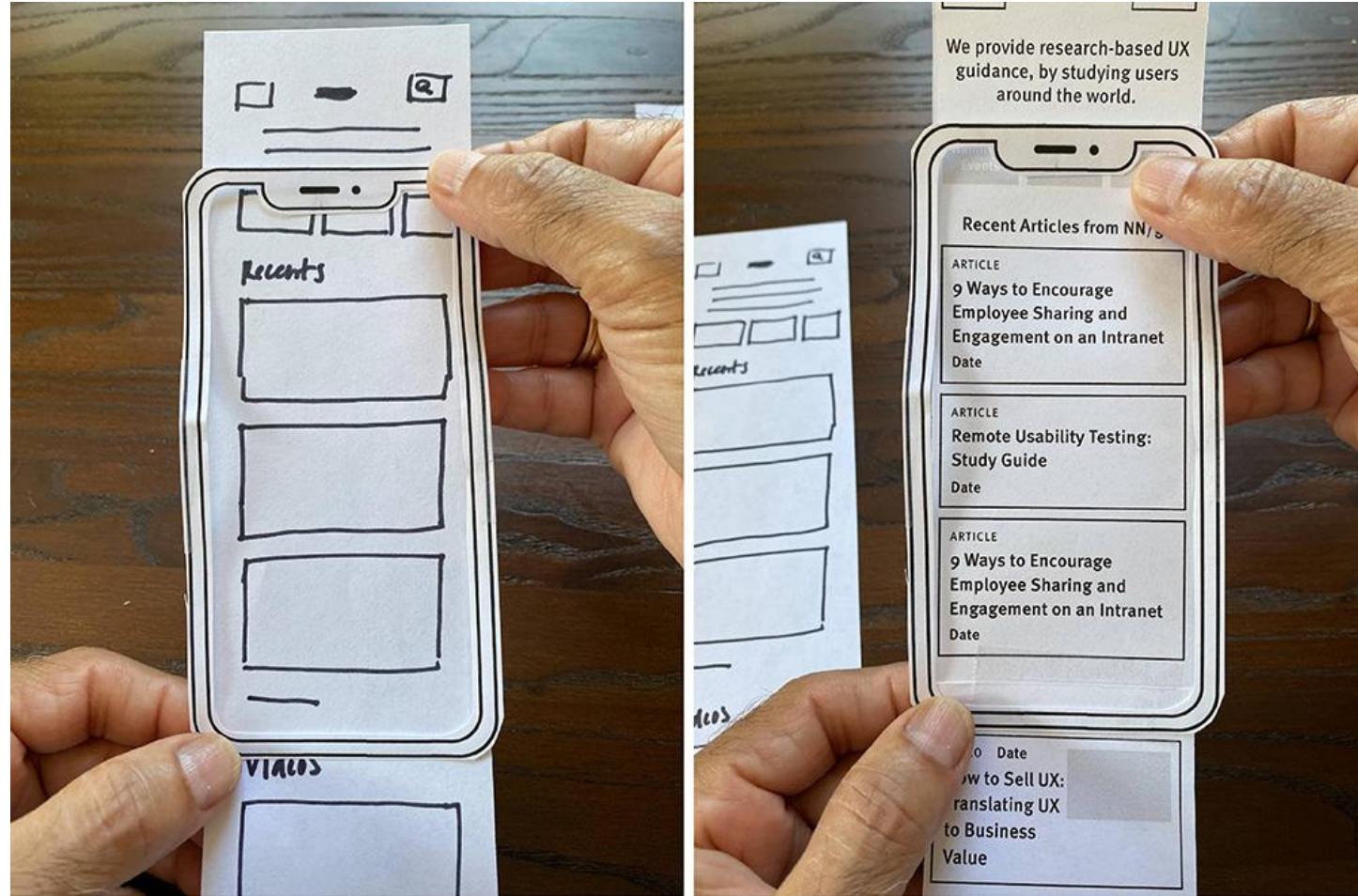


Here's some more text lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

10 Children [Show Children](#) | [View in Hierarchy](#) | [Add Child Page](#)

Paper Prototyping

- Paper prototyping is the process of sketching out potential concepts, flows or ideas on paper and testing those sketches with users



<https://www.nngroup.com/articles/paper-prototyping-cutout-kit/>

Paper Prototyping

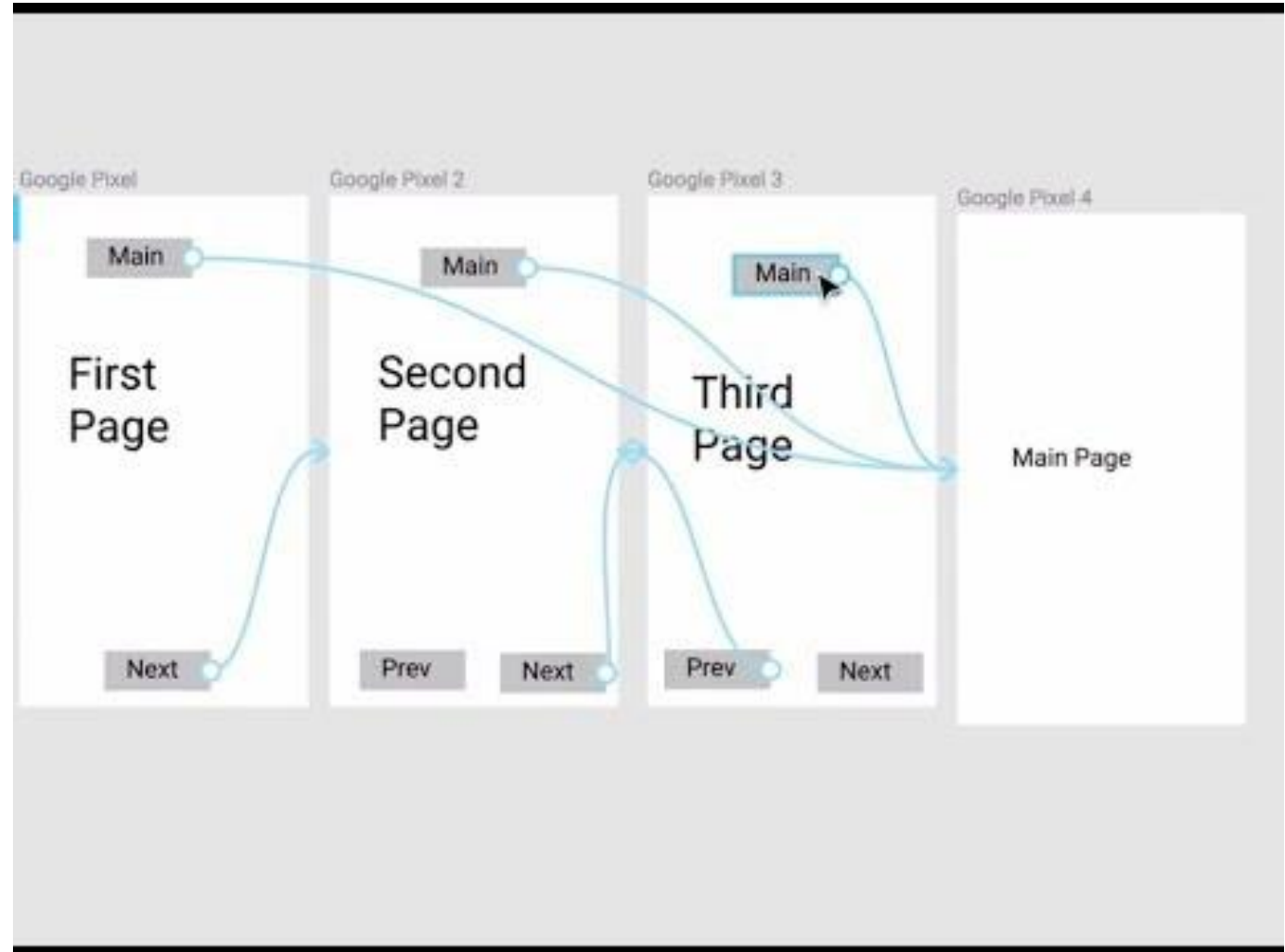
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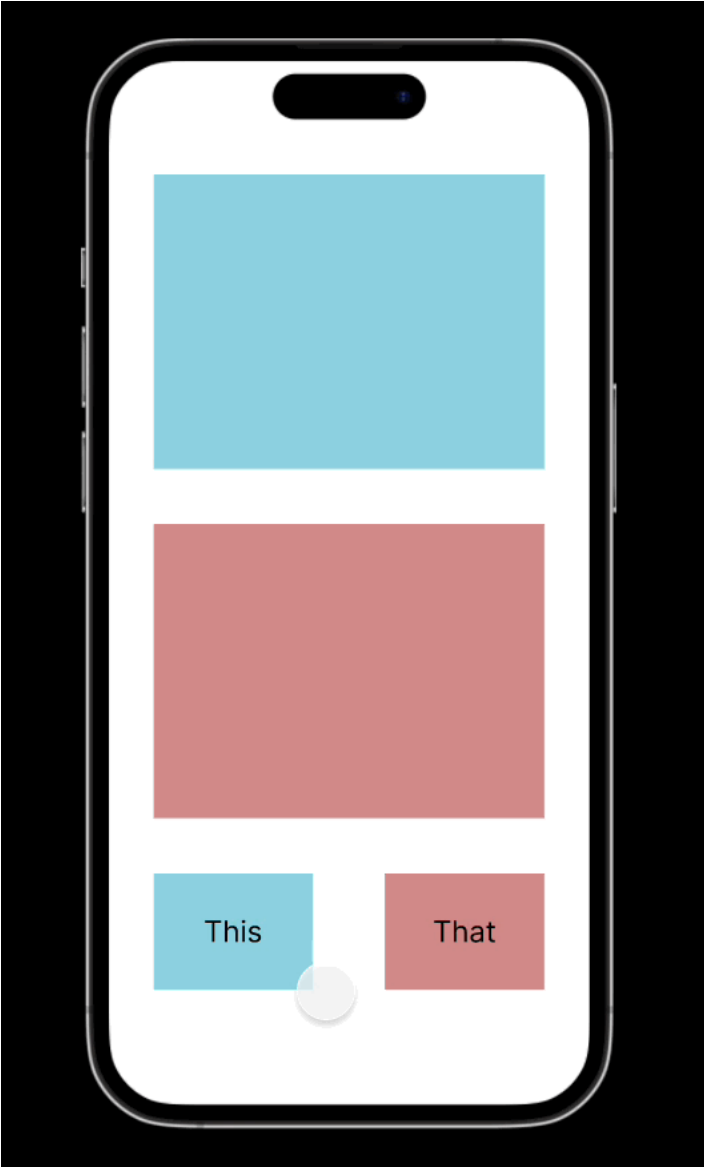
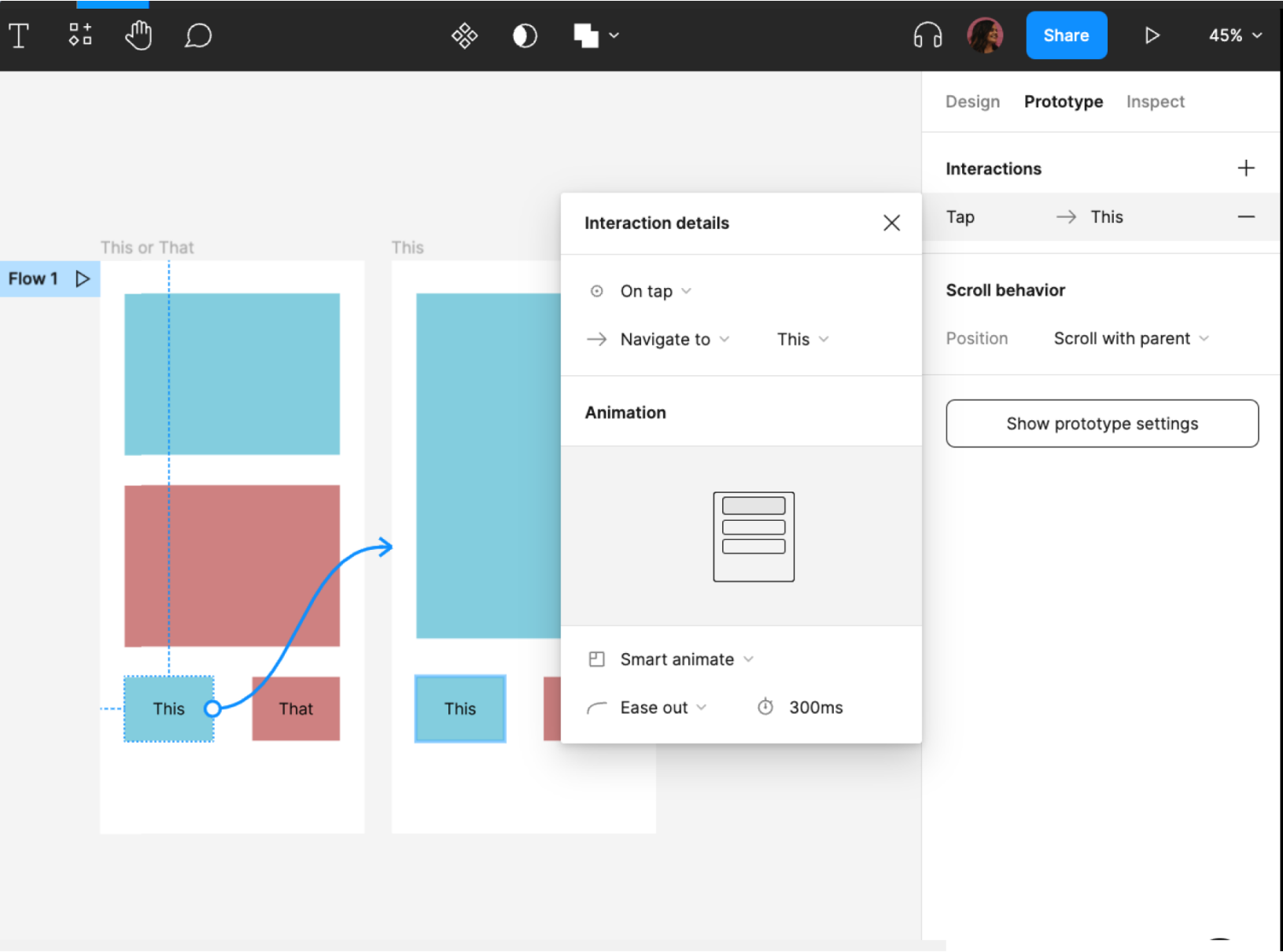


<https://martha-eierdanz.com/paper-prototyping-in-practice>

Click-through Prototypes

e.g., Figma

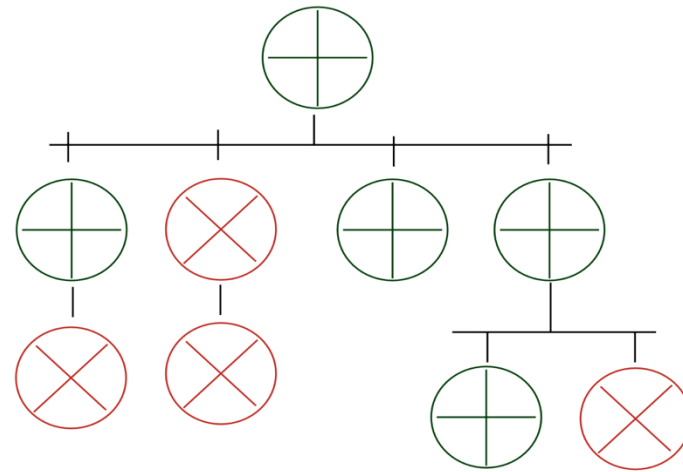




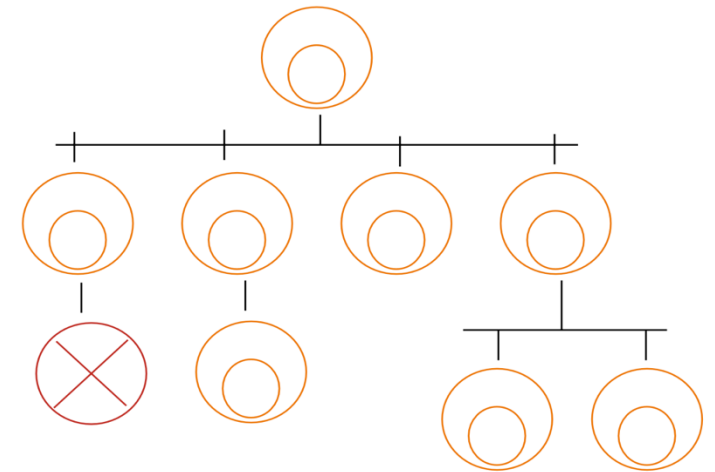
High-fidelity Prototypes

Types of high-fidelity prototypes

Vertical prototyping



Horizontal prototyping



A **horizontal** prototype is a prototype, where all the visual parts of the **user interface** of a new computer system is implemented, i.e. screen dialogues and their interconnections can be demonstrated, but no data can be processed. In contrast a **vertical** prototype is a prototype where **a few selected functions** are implemented in such detail that realistic data can be processed, i.e. a realistic work task can be performed with a vertical prototype. (Grønbaek, 1989)



Fully implemented



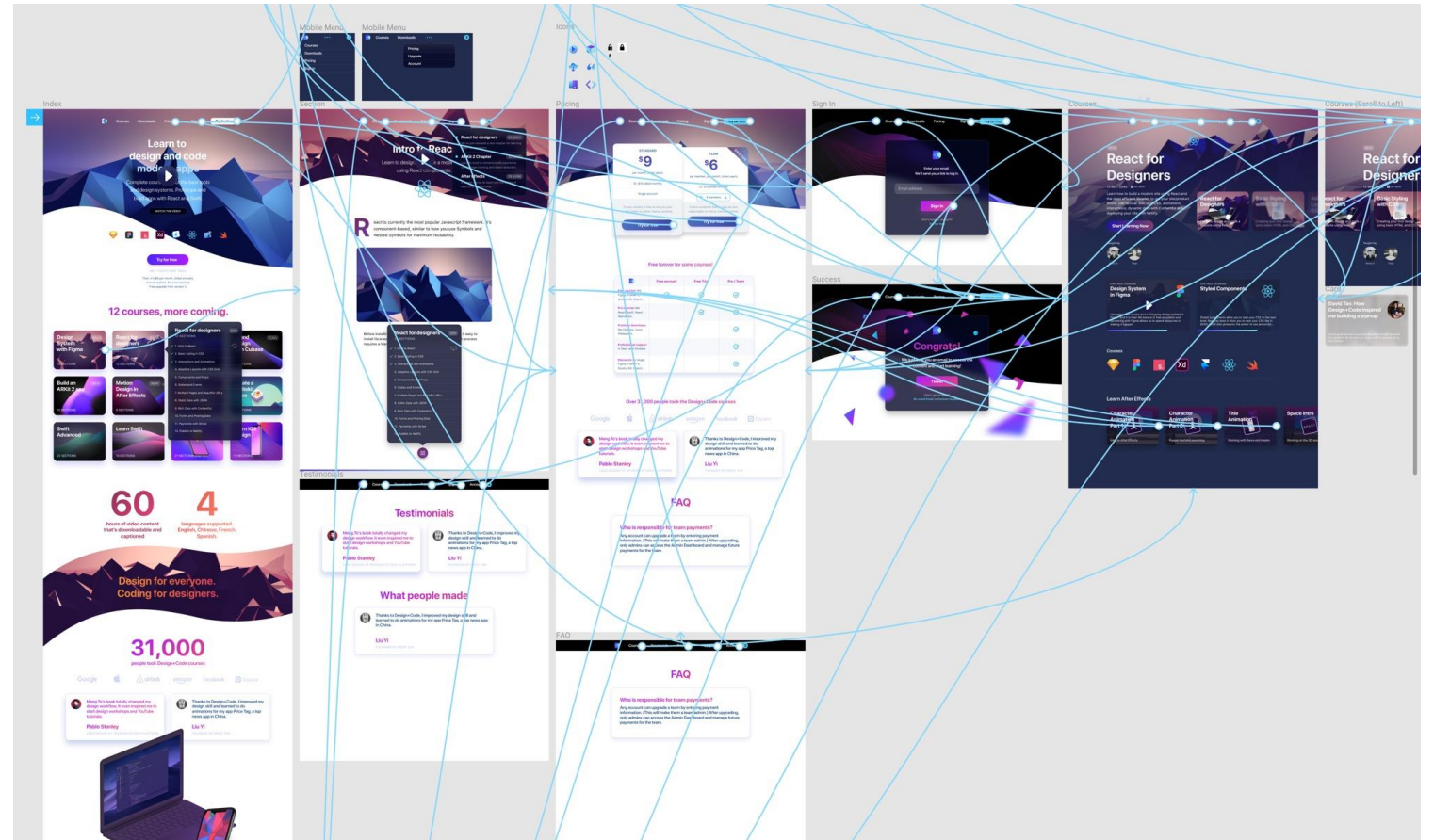
Partially implemented



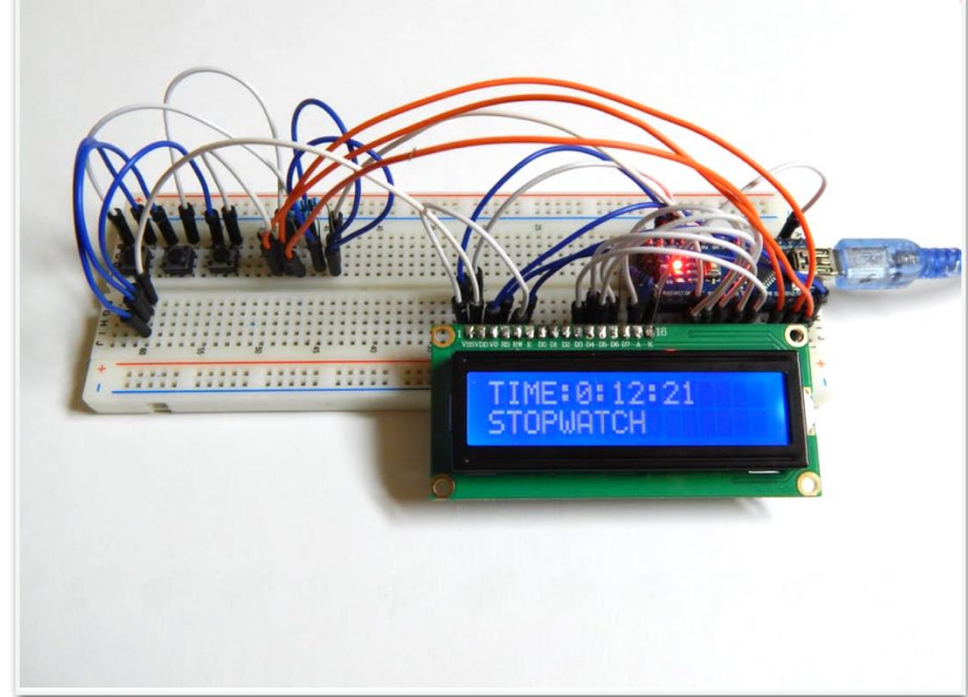
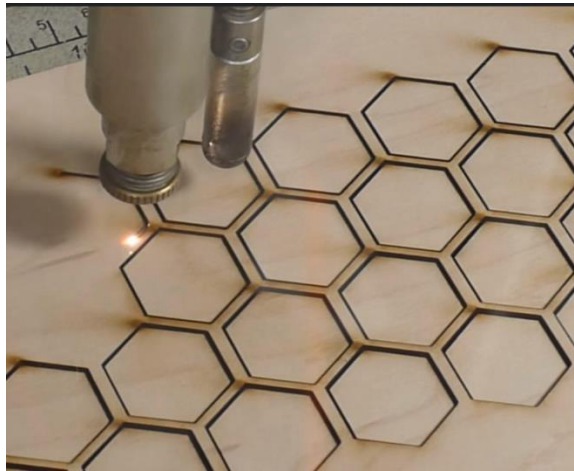
Not implemented

Click-through Prototypes

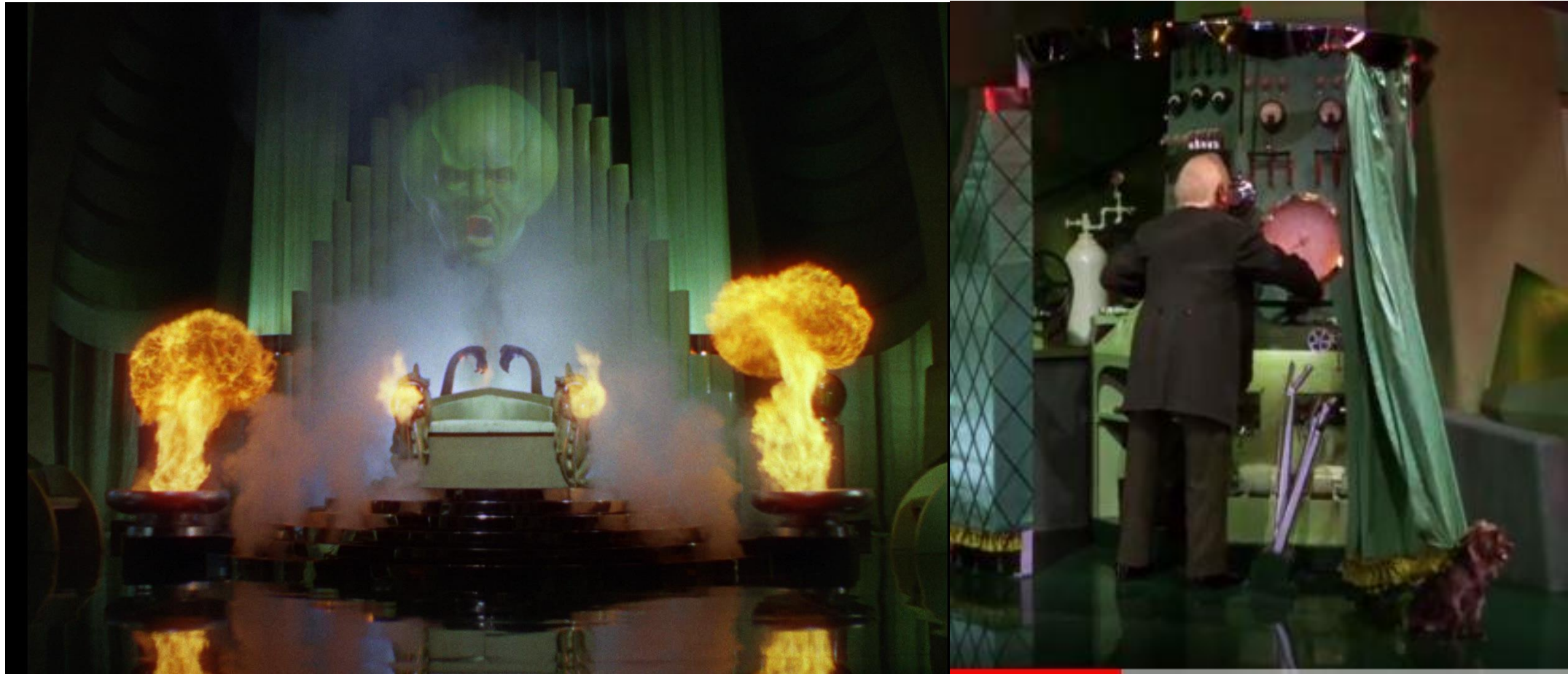
e.g., Figma



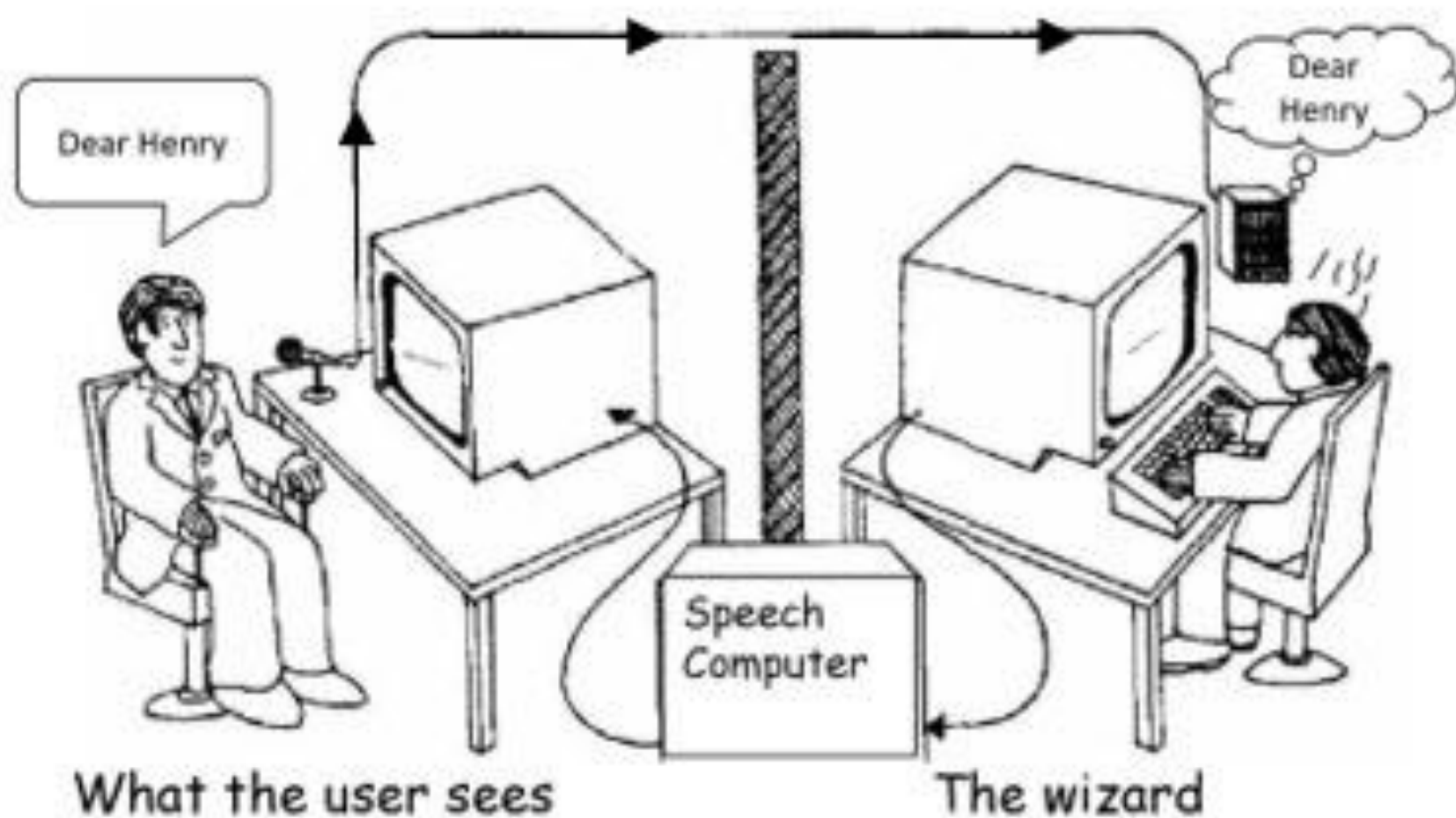
Physical Computing



Wizard of Oz Prototyping



Wizard of Oz testing – The listening type writer IBM 1984



AI Use Case

Recommender system

Chatbot

Classifier UI

Prototype Strategy

Mock interface with fake suggestions

Wizard-of-Oz with scripted responses

Confidence bars + explanation cards

Remember!

- what assumptions are your prototypes testing?
- what does this mean for how/why you prototype?

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